Facilitative Processing of Gender in the Absence of Overt Agreement Morphology in Spanish Heritage Speakers

USCPoMMLab
Psycholinguistics of Mono- & Multilingualism

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1. Introduction

- **Prior work**: Despite comparatively limited input than baseline Spanish speakers, HSs show facilitative processing of grammatical gender (ex. Fuchs, 2021; Pérez-Leroux et al., 2024; Keating, 2022)
- Grammatical gender & number are both lexical properties of Spanish nouns
- HSs use gender and number facilitatively with Spanish definite articles (Coloma et al., 2023)

Definite

[M.SG] Indica (Show) **el** (the.MASC) libro (book.MASC)

[M.PL] Indica (Show) **los** (the.MASC.PL) libros (book.MASC.PL)

- **HSs tend to diverge from the baseline in interpreting and producing linguistic constructions that contain morphologically null elements** (Laleko & Polinsky, 2017)
  - Production and comprehension of null vs. overt pronouns (Sorace & Serratrice, 2009)
- Morphosyntactic gender agreement in Russian: (Fuchs & Sekerina, 2025)
  - HSs exhibited facilitative use of F (two overt markings) but not M (one null & one overt marking)
    ⇒ Due to absence of overt morphology or due to distinctions between M vs. F as gender features?

Do HSs access gender & number information on a pre-nominal article in the absence of overt agreement morphology?

Motivation: pre-nominal material that inflects for gender in Spanish is limited

Indefinite

[M.SG] ¿Hay (Is there) **algún.Ø** (some.MASC) libro? (book.MASC)

	Singular	Plural
Masculine	algún.Ø	algun-os
Feminine	algun-a	algun-as

2. Methodology

Research Questions & Predictions

In the absence of overt agreement morphology, do HSs exhibit facilitative processing of...

1. Grammatical gender? → Gender-informative > Uninformative [M.SG]
2. Number? → Number-informative > Uninformative [M.SG]

Participants

HSs n = 32

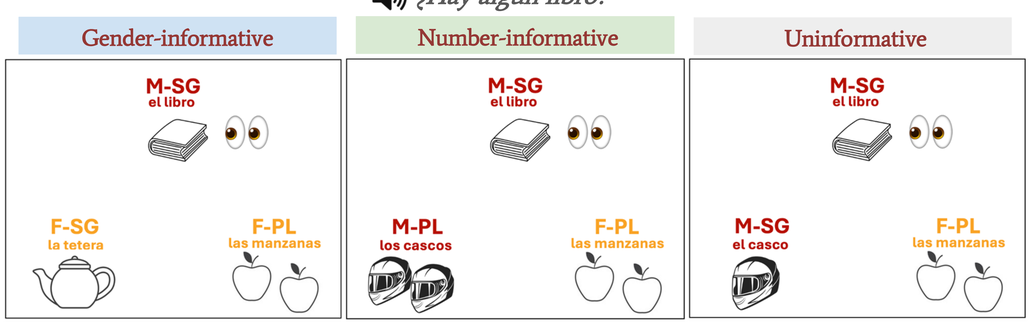
Age: mean = 19.8 yrs (sd = 1.86)
Age of arrival to US: mean = 4.75 yrs (sd = 1.1)

- Grew up speaking Spanish with one or both parents
- Born in a Spanish-speaking country & moved to US prior to age 6 {or} born in US

Eye-tracking in the Visual World Paradigm

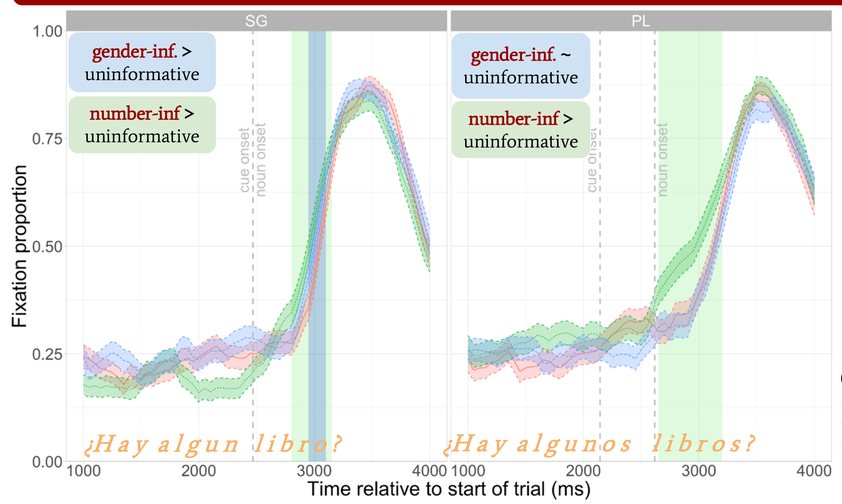
Auditory prompt: ¿Hay (Is there) { algun (SG), algun.os (PL) M; algun.a (SG), algun.as (PL) F } [NOUN(-PL)]?

¿Hay algun libro?



256 displays balanced for gender & number of items
32 nouns 16 masc. & 16 fem.
64 fillers for each list
2 lists

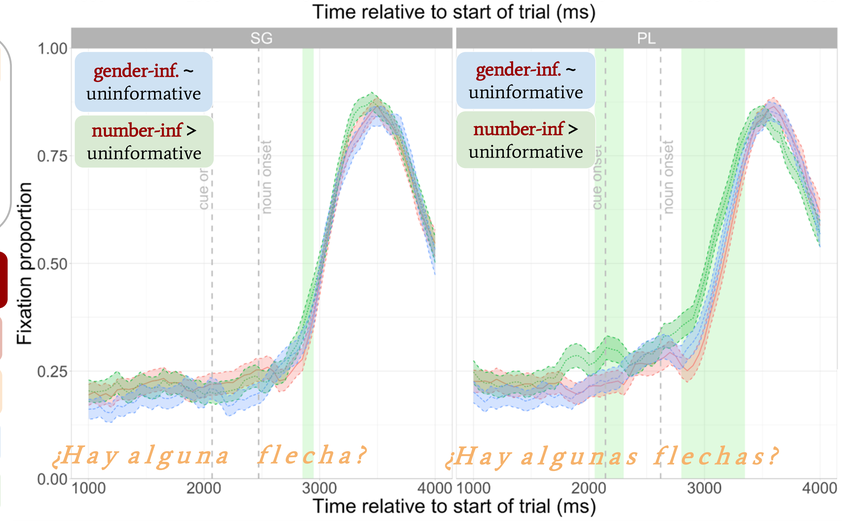
3. Results



MASC targets

Prediction	M.SG	M.PL
Gender informative > uninformative	cluster mass statistic = 18.5	no sig. cluster
Number informative > uninformative	cluster mass statistic = 88.0	cluster mass statistic = 150.

Ask me about sig. effects in other regions



FEM targets

Prediction	F.SG	F.PL
Gender informative > uninformative	no sig. cluster	no sig. cluster
Number informative > uninformative	cluster mass statistic = 15.2	cluster mass statistic = 150.

Ask me about sig. effects in other regions

4. Conclusions

Gender-informative > Uninformative [M.SG] → Heritage speakers show **facilitative processing of grammatical gender in the absence of overt agreement morphology.**

Number-informative > Uninformative → Heritage speakers show **facilitative processing of number in the absence of overt agreement morphology.**

Number effects across all conditions; gender effects only in [M.SG] → In line with the bilingualism literature: non-dominant **bilingual speakers tend to prioritize semantically salient cues over abstract grammatical ones** (Clahsen & Felser, 2018; Cunnings, 2017; Parshina et al., 2022)

Do HSs access gender & number information on a pre-nominal article in the absence of overt morphology? → YES → Despite limited input, heritage speakers show **facilitative processing even with a lower frequency construction *algún***

Outstanding puzzle: Why do HSs **exhibit facilitative processing of gender** with null **M.SG** but not with overt forms in **M.PL**, **F.SG** & **F.PL**?
Vowel duration (longer with M.SG) as a cue?

Selected References: Cunnings, I. (2017). Interference in native and non-native sentence processing. **Fuchs**, Z. (2021). Facilitative use of grammatical gender in Heritage Spanish. **Fuchs**, Z., and Sekerina, I. A. (2025). New evidence for the role of morphological markedness of gender agreement cues in monolingual and heritage-bilingual facilitative processing. **Keating**, G. D. (2022). The effect of age of onset of bilingualism on gender agreement processing in Spanish as a heritage language. **Ito**, A., Nguyen, H. T. T., & Knoeferle, P. (2023). German-dominant Vietnamese heritage speakers use semantic constraints of German for anticipation during comprehension in Vietnamese. **Laleko**, O & Polinsky, M. (2017). Silence is difficult: On missing elements in bilingual grammars.. **Parshina**, O., Lopukhina, A., and Sekerina, I. A. (2022). Can heritage speakers predict lexical and morphosyntactic information in reading? **Pérez-Leroux**, A. T., Colantoni, L., Thomas, D., and Chen, C. H. (2024). The morphophonological dimensions of Spanish gender marking: NP processing in Spanish bilinguals.